External Data Providers & APIs
Yahoo Finance (yfinance) Daily OHLCV, FastInfo, Metrics, VIX
NSE Official Archive EQUITY_L.csv & Ind Lists
BSE Disclosures & IR Portal Corporate Announcements
RBI & Macro Official Releases Repo Rate, CPI, GDP, 10Y Yield
Google Gemini API (gemini-3.8-flash via LangChain)
Transformation, Validation & Storage Layer
Data Quality & Freshness Gate (<= 14-day quotes, <= 7-day fundamentals)
PostgreSQL ACID Storage (62 Tables across 8 Domains) securities, market_prices, fundamental_metrics, economic_observations, investor_portfolios
Candidate Universe Selector Filters 2,639 Master -> ~504 Recommendation Pool
Risk Assessment
Quality Narrative
Momentum Reasoning
Qualitative Synthesis
Governance Audit
Strategy Synthesis
Fiduciary Explanations
LangGraph Shared State (PortfolioAgentState)
PortfolioAgentState Payload - user_profile & financial_goals - eligible_candidates (~504) - macro_output & market_regime - fundamental_factors & news_audits - proposed_allocations & critic_findings
LangGraph Multi-Agent Research Committee
1. Manager Node Initializes Run & State Traces
2. Suitability Node Evaluates Risk Capacity & Feasibility
3. Macro Analyst Node India VIX, GDP, Repo Rate Stance
4. Market Analyst Node Nifty 50 Trend, 50/200 DMA, Volatility
5. Fundamental Analyst Node ROE > 18%, P/E, Debt/Equity, Cash Flow
6. News & Filings Analyst Node BSE/NSE Announcements, Forensic Scrutiny
7. Lead Strategist Node Selects Top 8-15 Assets & Factor Tilts
8. Independent Risk Critic Node Strict Fiduciary Audit & Single-Stock <= 25% Ceiling
Rework Required
9. Bounded Rework Node (Max 2 iterations if critique fails)
Approved
Deterministic Math & Output Phase
10. Deterministic SciPy SLSQP Optimizer Zero-LLM Influence; Integer Whole Shares; Unallocated Cash Conservation
11. Critic Final Validation Post-Optimization Allocation & Boundary Audit
FailedPassed
12. Fiduciary Moderator Generates Portfolio & Stock-Level AI Rationale
13. ACID PostgreSQL Persistence Commits Portfolio Version & Decisions
Investor Web Application (Next.js / React) 'My Portfolio' Tab, 'Why' Modals, Rebalancing